Andre Arante

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Education

The George Washington University Washington, DC
Bachelor of Science in Electrical and Computer Engineering *Aug 2023 – May 2027*

- Minor in Computer Science; coursework in Computer Vision, Reinforcement Learning, and Machine Learning

Publications

1. **A. Arante**, M. Ghazanfari, C. Vu, and P. Wei, "Open-Vocabulary Obstruction Detection for UAS Landing via Fine-Tuned YOLO-World" submitted to *AIAA SciTech 2027*, under review.

Work Experience

Software Engineering Intern May 2026 – Present
Dell Technologies *Austin, TX*

- Built the physical execution layer for an agentic AI system by finetuning vision-language-action models (VLAs, such as π_0 and SmolVLA) via LoRA to translate natural language instructions into contact-rich robotic manipulation (keyboard macros, USB insertion) for autonomous hardware validation
- Extended the pipeline with HIL-SERL, training a custom residual RL policy that learns corrective actions ontop of a frozen VLA from real-time human feedback, improving task success

Undergraduate Research Intern Sep 2025 – Present
The George Washington University *Washington, DC*

- Ran experiments on the tradeoff between domain fine-tuning and open-vocabulary retention for open-vocabulary object detection models such as Yolo World and deployed for real-time landing zone obstruction detection for package delivery UAS drones in urban enviornments.
- Built autonomous driving test environments using CARLA/SUMO to model highway merging scenarios

Software Engineering Intern Feb 2024 – Feb 2025
C-3 Communication Systems, LLC *Bethesda, MD*

- Owned end-to-end development of an autonomous UGV platform, architecting a modular software stack that integrated computer vision, obstacle detection, and autonomous navigation into a unified control system.
- Shipped and maintained a full-stack Operations Management System (HTML/CSS/JS, Firebase) in production for 1+ year, automating payroll, time tracking, and logistics workflows for a small startup

Data Science Intern Jun 2023 – Aug 2023
Kaiser Permanente *San Jose, CA*

- Refactored a SAS data pipeline into an automated daily job processing millions of patient records across 5 hospitals and generated informative Tableau dashboards, eliminating a manual workflow that gated critical staffing decisions

Projects

SortBots – Distributed Multi-Robot Package Logistics (Senior Capstone)

- One of only 3 ECE student-proposed capstones selected from 19 submissions; only capstone project to independently secure an external sponsor to fully fund my project idea
- Designing a decentralized multi-robot coordination system for autonomous package logistics in warehouse environments using decentralized coordination, 3D perception (VSLAM), and multi-agent path planning

ImitationArm - Custom 6-DOF Robotic Arm Manipulator

- Developed a ROS 2 / C++ control stack integrating MoveIt2 for motion planning and MuJoCo for sim-to-real RL
- Built a custom wearable IMU-based teleoperation controller (ESP32 + MPU-6050) for real-time teleoperation

Reinforcement Learning Agents in Minecraft

- Trained agents via Behavioral Cloning from human demonstrations using custom Python scripts to record Minecraft gameplay, then fine-tuned with actor-critic networks (PPO) to preform Minecraft parkour and bridging tasks

Technical Skills

ML & Foundation Models: PyTorch, CUDA, fine-tuning (SFT, RL, LORA), Weights & Biases, vision-language models (CLIP, YOLO-World), vision-language-action models (π_0, SmolVLA)
Robotics & Deployment: C/C++, Python, ROS 2, MuJoCo, OpenCV, NVIDIA Jetson, TensorRT, Linux, Git, Docker